

FEA coursework

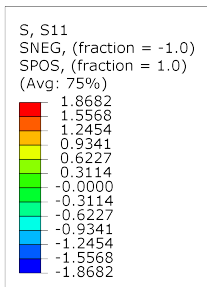
Celestine Dal

April 2025

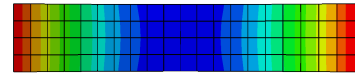
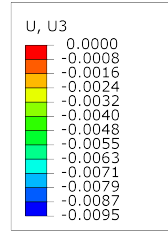
- Nlgeom = Non-linear geometry
- p = Distributed load per unit length
- L = Beam length
- M_b = Bending moment
- σ_x = Longitudinal stress (S11)
- δ = Displacement (U3)

1 Non-linear geometry questions

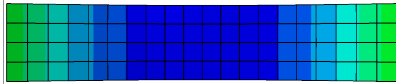
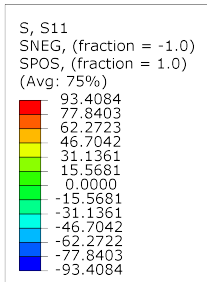
1.1 Varying the distributed load



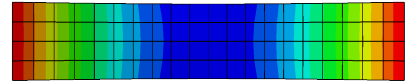
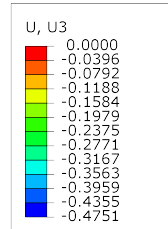
σ_x for Nlgeom off, $p = 0.02$ N/mm, $L = 100$ mm



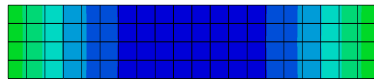
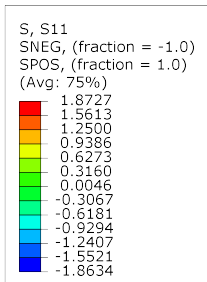
δ for for Nlgeom off, $p = 0.02$ N/mm, $L = 100$ mm



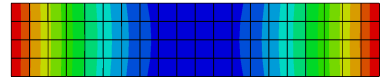
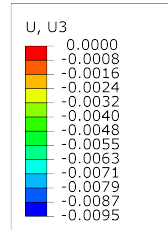
σ_x for Nlgeom off, $p = 1$ N/mm, $L = 100$ mm



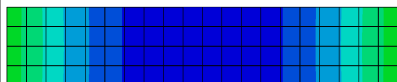
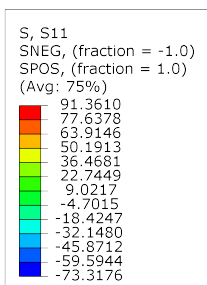
δ for for Nlgeom off, $p = 1$ N/mm, $L = 100$ mm



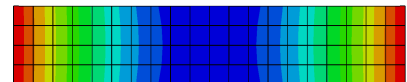
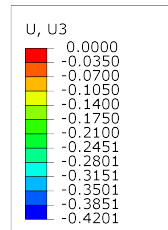
σ_x for Nlgeom on, $p = 0.02$ N/mm, $L = 100$ mm



δ for for Nlgeom on, $p = 0.02$ N/mm, $L = 100$ mm



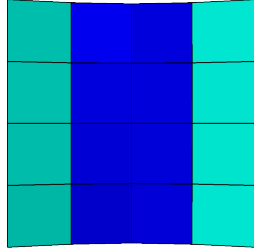
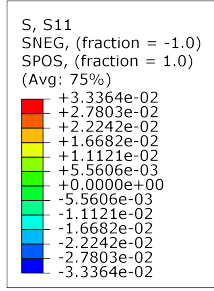
σ_x for Nlgeom on, $p = 1$ N/mm, $L = 100$ mm



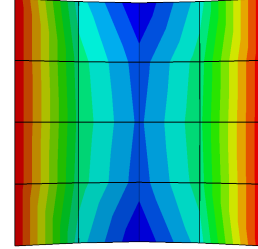
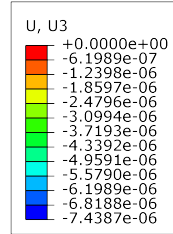
δ for for Nlgeom on, $p = 1$ N/mm, $L = 100$ mm

The results for displacement for beams subjected to very low loads can be seen to be identical for linear and non-linear geometries, however, as soon as the load is increased, the magnitude of the maximum displacement for the linear solution (0.4751 mm) is seen to be slightly greater than that of the non-linear solution (0.4201 mm), although still negligible. Increasing the load from 0.02 to 1 N/mm increases the peak stress by over 50 times.

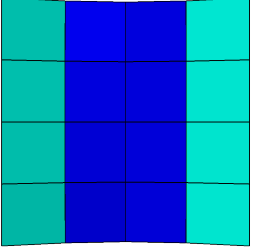
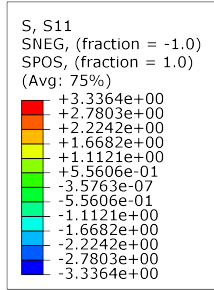
1.2 Varying the length of the beam



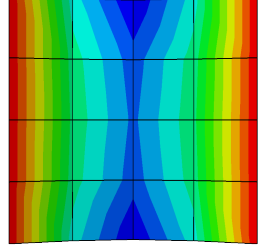
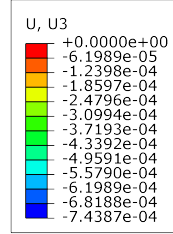
σ_x for Nlgeom off, $p = 0.01$ N/mm, $L = 20$ mm



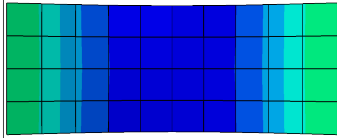
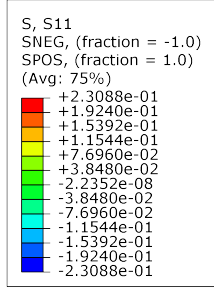
δ for for Nlgeom off, $p = 0.01$ N/mm, $L = 20$ mm



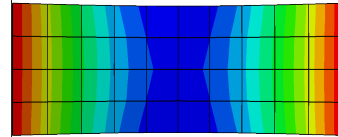
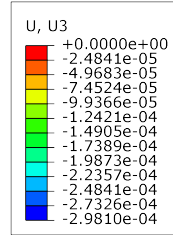
σ_x for Nlgeom off, $p = 1$ N/mm, $L = 20$ mm



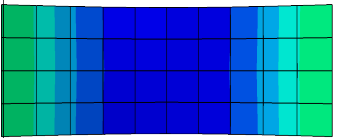
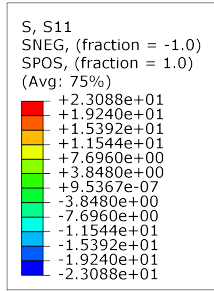
δ for for Nlgeom off, $p = 1$ N/mm, $L = 20$ mm



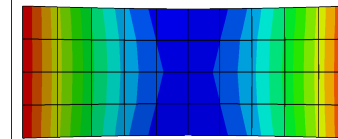
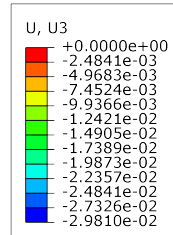
σ_x for Nlgeom off, $p = 0.01$ N/mm, $L = 50$ mm



δ for for Nlgeom off, $p = 0.01$ N/mm, $L = 50$ mm

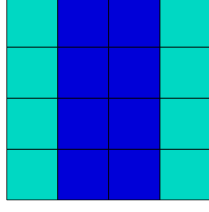
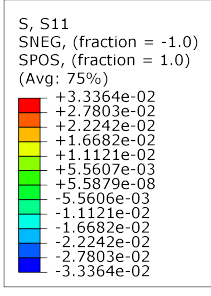


σ_x for Nlgeom off, $p = 1$ N/mm, $L = 50$ mm

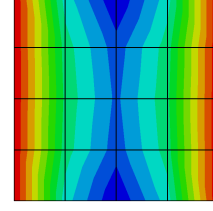
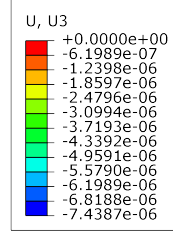


δ for for Nlgeom off, $p = 1$ N/mm, $L = 50$ mm

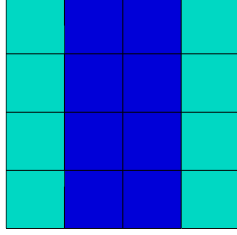
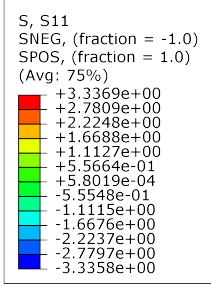
For the linear solutions, it can be seen above that an increase in beam length corresponds to an increase in stress and an increase in magnitude of displacement, both by a factor $\times 10^2$.



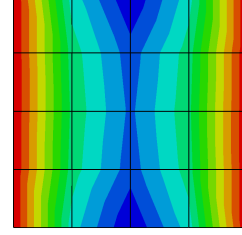
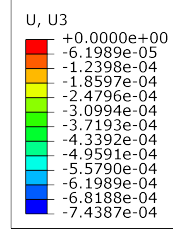
σ_x for Nlgeom on, $p = 0.01$ N/mm, $L = 20$ mm



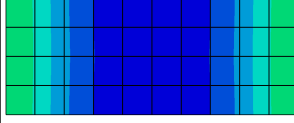
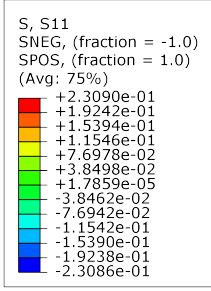
δ for for Nlgeom on, $p = 0.01$ N/mm, $L = 20$ mm



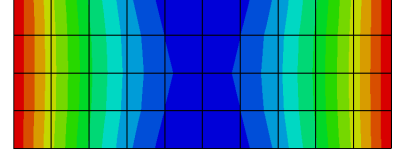
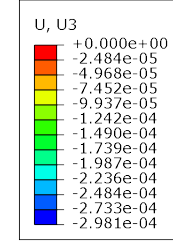
σ_x for Nlgeom on, $p = 1$ N/mm, $L = 20$ mm



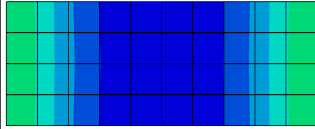
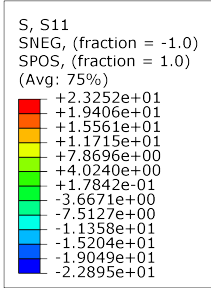
δ for for Nlgeom on, $p = 1$ N/mm, $L = 20$ mm



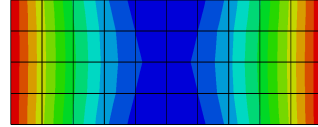
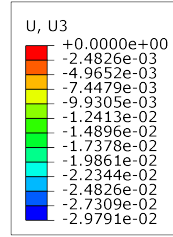
σ_x for Nlgeom on, $p = 0.01$ N/mm, $L = 50$ mm



δ for for Nlgeom on, $p = 0.01$ N/mm, $L = 50$ mm



σ_x for Nlgeom on, $p = 1$ N/mm, $L = 50$ mm

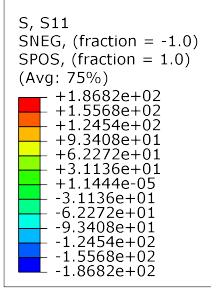


δ for for Nlgeom on, $p = 1$ N/mm, $L = 50$ mm

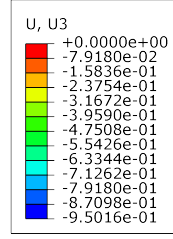
For the non-linear solutions, it can be seen above that an increase in beam length corresponds to an increase in stress and an increase in magnitude of displacement, both by a factor $\times 10^2$.

For both the linear and non-linear solutions, the effects of varying the load and length of beam produce very similar results and the differences can be considered to be negligible. Nlgeom has a greater impact on longer beams subjected to a greater load because they are less stiff, flexible and prone to bending.

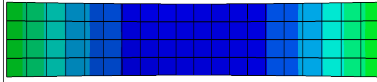
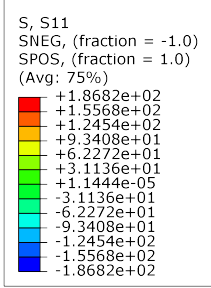
1.3 Hinged vs simply supported beam



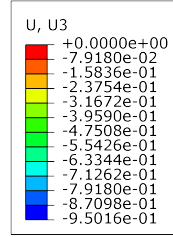
σ_x for Nlgeom off, $p = 2$ N/mm, $L = 100$ mm
(hinge-hinge)



δ for for Nlgeom off, $p = 2$ N/mm, $L = 100$ mm
(hinge-hinge)

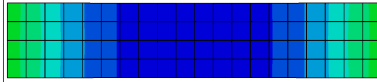
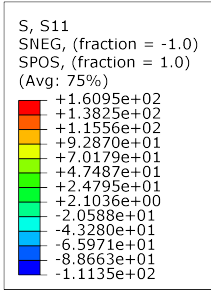


σ_x for Nlgeom off, $p = 2$ N/mm, $L = 100$ mm
(simply-supported)

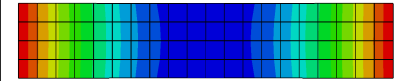
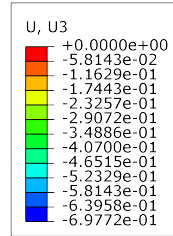


δ for for Nlgeom off, $p = 2$ N/mm, $L = 100$ mm
(simply-supported)

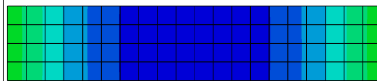
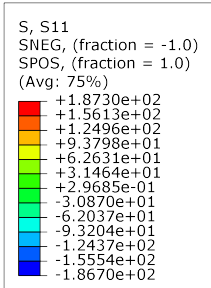
For linear solutions, it can be seen above that the stress and displacements stayed the same for both hinge-hinge and simply-supported beams because it is assumed that the beam does not stretch.



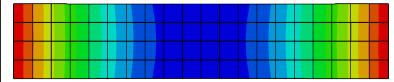
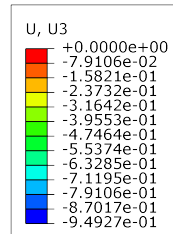
σ_x for Nlgeom on, $p = 2$ N/mm, $L = 100$ mm
(hinge-hinge)



δ for for Nlgeom on, $p = 2$ N/mm, $L = 100$ mm
(hinge-hinge)



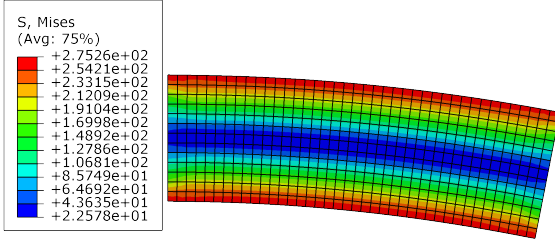
σ_x for Nlgeom on, $p = 2$ N/mm, $L = 100$ mm
(simply-supported)



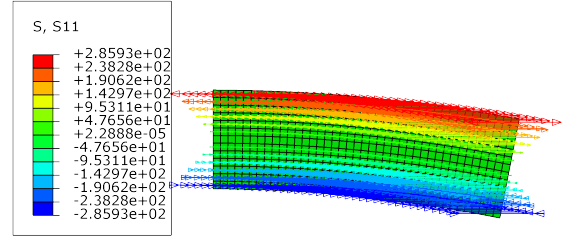
δ for for Nlgeom on, $p = 2$ N/mm, $L = 100$ mm
(simply-supported)

For non-linear solutions, the type of support on the beams greatly affected both stress and displacement, with much higher magnitudes for simply supported beams. This is because beam deformation is taken into account and this creates in-plane tension. This change in shape affects the stiffness and load paths, leading stress and displacement to increase for a simply-supported beam compared to a beam supported by just hinges.

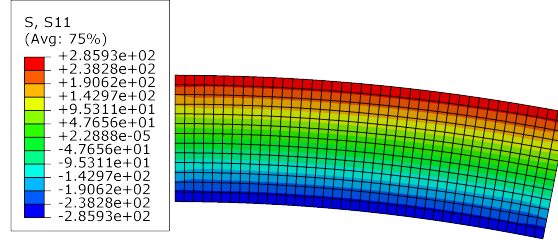
2 Material non-linearity questions



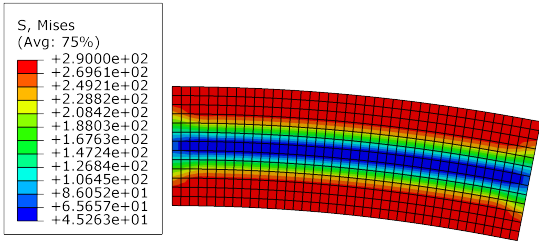
Von Mises for Nlgeom off, $M_b = 193333 \text{ Nmm}$



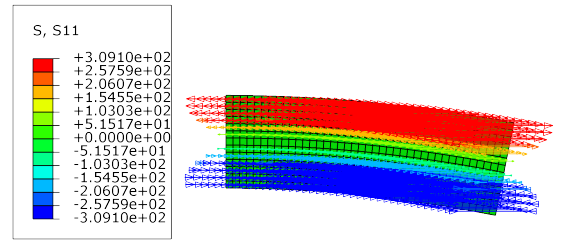
Deformation for Nlgeom off, $M_b = 193333 \text{ Nmm}$



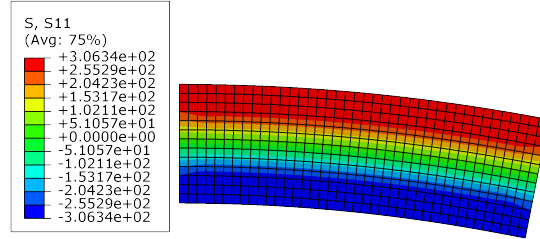
σ_x for Nlgeom off, $M_b = 193333 \text{ Nmm}$



Von Mises for Nlgeom off, $M_b = 265833 \text{ Nmm}$

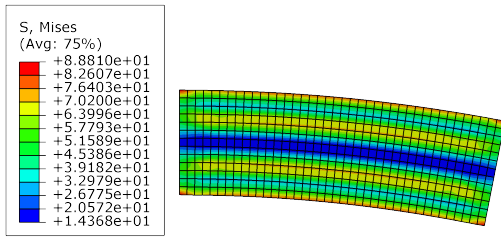


Deformation for Nlgeom off, $M_b = 265833 \text{ Nmm}$

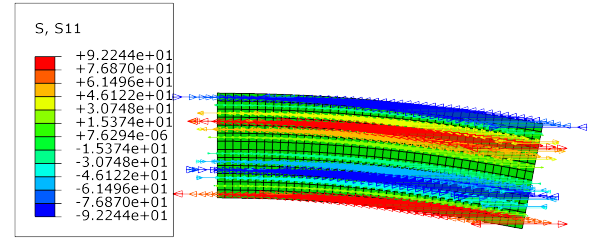


σ_x for Nlgeom off, $M_b = 265833 \text{ Nmm}$

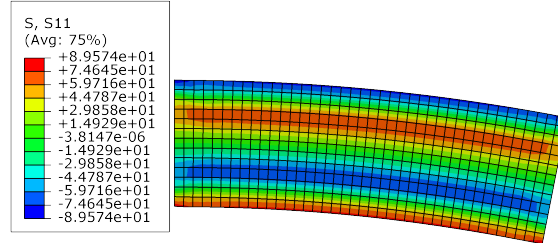
For elastic deformation, it is clear that an increase in bending moment applied to the beam increases the stress and magnitude of deformation of the beam. Increasing the bending moment by a factor of 1.375 led to an increase in σ_x by a factor of 1.071. The deformation diagram is also visibly more exaggerated for a higher bending moment, since plastic deformation is occurring.



Von Mises for Nlgeom off, after plastic deformation
 $(M_b = 265833 \rightarrow 0 \text{ Nmm})$



Deformation for Nlgeom off, after plastic deformation
 $(M_b = 265833 \rightarrow 0 \text{ Nmm})$



σ_x for Nlgeom off, after plastic deformation
 $(M_b = 265833 \rightarrow 0 \text{ Nmm})$

These diagrams show that once the bending moment has been fully removed, the beam does not go back to its original shape, as it has permanently deformed plastically. The diagram for σ_x shows the top internal core has plastically deformed but the bottom internal core has elastically deformed. After unloading, this difference in stresses leads to residual stress.